

Warm-up

For each integral, write the correct trig substitution:

$$(1) \int \frac{x^3}{\sqrt{4+x^2}} dx, \quad (2) \int \frac{1}{x^2\sqrt{9-x^2}} dx, \quad (3) \int \frac{\sqrt{1+9x^2}}{x^2} dx.$$

Medium

(1) Compute the integral (1) from the Warm-up section.

(2) Compute the integral (2) from the Warm-up section.

Evaluate the following integrals:

$$(3) \int_0^1 \frac{x^2}{\sqrt{4-x^2}} dx, \quad (4) \int_3^6 \frac{\sqrt{x^2-9}}{x} dx, \quad (5) \int e^x \sqrt{36-e^{2x}} dx.$$

Stretch

(1) Compute $\int \frac{\sqrt{1+9x^2}}{x^2} dx$, number (3) from the Warm-up section.

(2) Compute $\int (x-1)^3 \sqrt{2x-x^2} dx$.